Practice Questions for 2024-12-03-Duality

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1 Questions

1. Given the primal problem (P) and its dual (D) as shown in Slide 1, if the primal problem is feasible and bounded, what can we conclude about the dual problem?

   a. The dual problem is infeasible.

   b. The dual problem is unbounded.

   c. The dual problem is feasible and bounded, and the optimal values of the primal and dual objective functions are equal.

   d. The dual problem is feasible and bounded, but the optimal values of the primal and dual objective functions may differ.

   e. We cannot conclude anything about the dual problem's feasibility or boundedness.

2. Referring to Slide 2, explain the significance of the complementary slackness conditions $x_j z_j = 0$ and $y_i w_i = 0$ in relation to the optimal solutions of the primal and dual problems.

   a. These conditions are necessary but not sufficient for optimality.

   b. These conditions are sufficient but not necessary for optimality.

   c. These conditions are both necessary and sufficient for optimality, provided both primal and dual problems are feasible.

   d. These conditions are irrelevant to the optimality of the primal and dual solutions.

   e. These conditions only apply to unbounded linear programs.

3. Consider the graphical method illustrated in Slides 3 and 4. How does the optimal solution relate to the feasible region and the objective function?

   a. The optimal solution is any point within the feasible region.

   b. The optimal solution is the point in the feasible region that minimizes the objective function.

   c. The optimal solution is the point on the boundary of the feasible region that maximizes the objective function.

   d. The optimal solution is the centroid of the feasible region.

   e. The optimal solution is the point where the objective function line intersects the origin.

4. Slides 11-15 discuss sensitivity analysis. If only the resource vector b changes in the primal problem (P), how does this affect the optimal objective function value ζ^*, and what is the economic interpretation of the vector y^* in this context?

   a. ζ^* remains unchanged; y^* represents the reduced costs.

   b. The change in ζ^* is given by $(\Delta b)^T x^*$; y^* represents shadow prices.

   c. The change in ζ^* is given by $(\Delta c)^T y^*$; y^* represents shadow prices.

   d. The change in ζ^* is given by $(\Delta b)^T y^*$; y^* represents shadow prices.

   e. We cannot determine the change in ζ^* without knowing the change in c.

5. a. Given the primal problem (P) and its dual (D) as shown in Slide 1, if the primal problem is feasible and bounded, what can we conclude about the dual problem?

      A. The dual problem is infeasible.

      B. The dual problem is unbounded.

      C. The dual problem is feasible and bounded.

      D. The dual problem is either infeasible or unbounded.

      E. There is no definitive conclusion about the dual problem.

   b. Referring to Slide 2, explain the significance of the complementary slackness conditions $x_j z_j = 0$ and $y_i w_i = 0$ in relation to the optimal solutions of the primal and dual problems.

      A. They ensure that the primal and dual objective function values are equal at optimality.

      B. They guarantee the feasibility of both the primal and dual problems.

      C. They indicate which constraints and variables are binding at the optimal solution.

      D. They are necessary but not sufficient conditions for optimality.

      E. They are only relevant for problems with equality constraints.

   c. Consider the graphical method illustrated in Slides 3 and 4. How does the optimal solution relate to the feasible region and the objective function?

      A. The optimal solution is any point within the feasible region.

      B. The optimal solution is the point in the feasible region that minimizes the objective function.

      C. The optimal solution is the point on the boundary of the feasible region that maximizes the objective function.

      D. The optimal solution is the centroid of the feasible region.

      E. The optimal solution is the point where all constraints are binding.

6. Given the primal problem (P) and its dual (D) as shown in Slide 1, if the primal problem is feasible and bounded, what can we conclude about the dual problem?

   a. The dual problem is infeasible.

   b. The dual problem is unbounded.

   c. The dual problem is feasible and bounded, and its optimal value is equal to the primal optimal value.

   d. The dual problem is feasible but may be unbounded.

e. The dual problem may be infeasible or unbounded.

7. Referring to Slide 2, explain the significance of the complementary slackness conditions $x_j z_j = 0$ and $y_i w_i = 0$ in relation to the optimal solutions of the primal and dual problems.

   a. These conditions are necessary but not sufficient for optimality.

   b. These conditions are sufficient but not necessary for optimality.

   c. These conditions are both necessary and sufficient for optimality, provided both primal and dual problems are feasible.

   d. These conditions are irrelevant to the optimality of the solutions.

   e. These conditions only apply to unbounded problems.

8. Consider the graphical method illustrated in Slides 3 and 4. How does the optimal solution relate to the feasible region and the objective function?

   a. The optimal solution is any point within the feasible region.

   b. The optimal solution is the point in the feasible region that minimizes the objective function.

   c. The optimal solution is the point on the boundary of the feasible region that maximizes the objective function.

   d. The optimal solution is the centroid of the feasible region.

   e. The optimal solution is the point where the objective function line intersects the origin.

9. Slides 11-15 discuss sensitivity analysis. If only the resource vector b changes in the primal problem (P), how does this affect the optimal objective function value ζ^*, and what is the economic interpretation of the vector y^* in this context?

   a. ζ^* remains unchanged; y^* represents the optimal primal solution.

   b. ζ^* changes proportionally to the change in b; y^* represents the reduced costs.

   c. ζ^* changes according to $\Delta\zeta^* = (\Delta b)^T y^*$; y^* represents shadow prices.

   d. ζ^* changes inversely to the change in b; y^* represents the dual slack variables.

   e. The effect on ζ^* is unpredictable; y^* has no economic interpretation.

10. a. Given the primal problem (P) and its dual (D) as shown in Slide 1, if the primal problem is feasible and bounded, what can we conclude about the dual problem?

       A. The dual problem is infeasible.

       B. The dual problem is unbounded.

       C. The dual problem is feasible and bounded.

       D. The dual problem is either infeasible or unbounded.

       E. There is no definitive conclusion about the dual problem.

    b. Referring to Slide 2, explain the significance of the complementary slackness conditions $x_j z_j = 0$ and $y_i w_i = 0$ in relation to the optimal solutions of the primal and dual problems.

A. They ensure that the primal and dual objective function values are equal at optimality.

B. They guarantee the existence of an optimal solution for both the primal and dual problems.

C. They indicate that at optimality, either a primal variable or its corresponding dual slack variable must be zero (or both).

D. They are necessary but not sufficient conditions for optimality.

E. They are only relevant for problems with equality constraints.

c. Consider the graphical method illustrated in Slides 3 and 4. How does the optimal solution relate to the feasible region and the objective function?

A. The optimal solution is any point within the feasible region.

B. The optimal solution is the point in the feasible region that minimizes the objective function.

C. The optimal solution is the point on the boundary of the feasible region that maximizes the objective function.

D. The optimal solution is the centroid of the feasible region.

E. The optimal solution is the point where all constraints are binding.

d. Slides 11-15 discuss sensitivity analysis. If only the resource vector b changes in the primal problem (P), how does this affect the optimal objective function value ζ^*, and what is the economic interpretation of the vector y^* in this context?

A. ζ^* remains unchanged; y^* represents the reduced costs.

B. ζ^* changes proportionally to the change in b; y^* represents the shadow prices.

C. ζ^* changes inversely to the change in b; y^* represents the dual slack variables.

D. ζ^* changes unpredictably; y^* represents the primal slack variables.

E. ζ^* remains unchanged; y^* represents the optimal primal solution.

2 Answers

1. Given the primal problem (P) and its dual (D) as shown in Slide 1, if the primal problem is feasible and bounded, what can we conclude about the dual problem?

   a. Incorrect. If the primal is feasible and bounded, the dual must also be feasible and bounded due to strong duality.

   b. Incorrect. A feasible and bounded primal problem implies a feasible and bounded dual problem.

   c. The strong duality theorem states that if the primal problem is feasible and bounded, then the dual problem is also feasible and bounded, and the optimal values of their objective functions are equal. This is a fundamental result in linear programming duality.

   d. Incorrect. The optimal values of the primal and dual objective functions are equal when both are feasible and bounded.

   e. Incorrect. Strong duality provides a definitive conclusion about the dual problem's properties.

2. Referring to Slide 2, explain the significance of the complementary slackness conditions $x_j z_j = 0$ and $y_i w_i = 0$ in relation to the optimal solutions of the primal and dual problems.

   a. Incorrect. While they are necessary, they are also sufficient for optimality given feasibility of both primal and dual.

   b. Incorrect. They are necessary conditions, and when combined with primal and dual feasibility, they are also sufficient.

   c. Complementary slackness conditions are necessary and sufficient for optimality when both the primal and dual problems are feasible. They state that at optimality, for each primal variable, either the variable is zero or its corresponding dual constraint is binding (and vice versa for dual variables and primal constraints).

   d. Incorrect. Complementary slackness is a fundamental concept in duality theory and crucial for determining optimality.

   e. Incorrect. Complementary slackness applies to both bounded and unbounded linear programs, but its interpretation might differ in the unbounded case.

3. Consider the graphical method illustrated in Slides 3 and 4. How does the optimal solution relate to the feasible region and the objective function?

   a. Incorrect. The optimal solution is not just any point; it's a specific point on the boundary.

   b. Incorrect. The description applies to minimization problems, not maximization.

   c. In the graphical method, the optimal solution for a maximization problem is always located at a vertex (or corner point) of the feasible region. This vertex is the point on the boundary of the feasible region that is furthest in the direction of improvement of the objective function.

   d. Incorrect. The centroid is not relevant to the optimal solution in linear programming.

   e. Incorrect. The intersection with the origin is irrelevant unless the origin is part of the feasible region.

4. Slides 11-15 discuss sensitivity analysis. If only the resource vector b changes in the primal problem (P), how does this affect the optimal objective function value ζ^*, and what is the economic interpretation of the vector y^* in this context?

a. Incorrect. A change in b will generally change ζ^*, and y^* represents shadow prices, not reduced costs.

b. Incorrect. The change in ζ^* due to a change in b is given by $(\Delta b)^T y^*$, not $(\Delta b)^T x^*$.

c. Incorrect. The change in ζ^* due to a change in b is given by $(\Delta b)^T y^*$, not $(\Delta c)^T y^*$.

d. When only the resource vector b changes, the change in the optimal objective function value is given by $\Delta\zeta^* = (\Delta b)^T y^*$. The optimal dual solution vector y^* represents the shadow prices, which are the marginal values of the resources. Each component of y^* indicates how much the optimal objective function value would change if one unit of the corresponding resource were added.

e. Incorrect. The change in ζ^* due to a change in b can be calculated using the optimal dual solution y^*, regardless of changes in c.

5. a. Given the primal problem (P) and its dual (D) as shown in Slide 1, if the primal problem is feasible and bounded, what can we conclude about the dual problem?

   A. Answer A is incorrect. The strong duality theorem states that if the primal is feasible and bounded, the dual is feasible and bounded. Infeasibility in the dual would imply unboundedness in the primal.

   B. Answer B is incorrect. An unbounded dual would imply an infeasible primal. Since the primal is given as feasible and bounded, the dual must also be bounded.

   C. The correct answer is C. If the primal problem (P) is feasible and bounded, then the dual problem (D) is also feasible and bounded. This is a direct consequence of the strong duality theorem in linear programming, which states that if both the primal and dual problems are feasible, then they both have optimal solutions, and the optimal objective function values are equal. Since (P) is bounded, its optimal value is finite. Strong duality guarantees that (D) also has a finite optimal value, implying it is bounded. The feasibility of (P) implies the feasibility of (D) under the conditions of strong duality.

   D. Answer D is incorrect. The strong duality theorem provides a more precise conclusion. The dual problem must be feasible and bounded.

   E. Answer E is incorrect. The strong duality theorem in linear programming provides a definitive conclusion about the dual problem's feasibility and boundedness given the primal problem's properties.

   b. Referring to Slide 2, explain the significance of the complementary slackness conditions $x_j z_j = 0$ and $y_i w_i = 0$ in relation to the optimal solutions of the primal and dual problems.

   A. Answer A is incorrect. While complementary slackness is related to the equality of primal and dual objective values at optimality, it doesn't directly ensure this equality. The equality is a consequence of strong duality.

   B. Answer B is incorrect. Feasibility is determined by the constraints of the primal and dual problems, not by complementary slackness. Complementary slackness is a condition that holds *at* optimality.

   C. The correct answer is C. The complementary slackness conditions $x_j z_j = 0$ and $y_i w_i = 0$ are crucial because they reveal which constraints and variables are binding (active) at the optimal solution. $x_j z_j = 0$ means that for each primal variable x_j, either $x_j = 0$ (non-binding) or its corresponding reduced cost $z_j = 0$ (binding). Similarly, $y_i w_i = 0$ means that for each dual variable y_i, either $y_i = 0$ (non-binding) or its corresponding primal slack variable $w_i = 0$ (binding). These conditions are essential for verifying optimality and for understanding the economic interpretation of the dual variables (shadow prices).

   D. Answer D is incorrect. Complementary slackness, along with primal and dual feasibility, is both necessary and sufficient for optimality.

E. Answer E is incorrect. Complementary slackness applies to problems with inequality constraints, where slack variables are introduced to convert inequalities to equalities. The conditions relate to the slack variables in the primal and dual problems.

c. Consider the graphical method illustrated in Slides 3 and 4. How does the optimal solution relate to the feasible region and the objective function?

A. Answer A is incorrect. The optimal solution is a specific point within the feasible region, not just any point.

B. Answer B is incorrect. The statement is correct for minimization problems, but the question refers to both maximization and minimization problems.

C. The correct answer is C. In the graphical method for linear programming, the feasible region is the set of all points that satisfy all the constraints. The objective function is a linear function that we want to maximize or minimize. The optimal solution is the point within the feasible region that yields the maximum (or minimum) value of the objective function. For linear programs, this optimal solution will always lie on a vertex (or corner point) of the feasible region, or along an edge if the objective function is parallel to a constraint.

D. Answer D is incorrect. The centroid is the average of all points in the feasible region, which is not necessarily the optimal solution.

E. Answer E is incorrect. While the optimal solution often lies at a point where multiple constraints are binding, it is not a requirement for all constraints to be binding at the optimum.

6. Given the primal problem (P) and its dual (D) as shown in Slide 1, if the primal problem is feasible and bounded, what can we conclude about the dual problem?

   a. Incorrect. If the primal is feasible and bounded, the dual must also be feasible and bounded by strong duality.

   b. Incorrect. A bounded primal problem implies a bounded dual problem due to strong duality.

   c. The strong duality theorem states that if the primal problem is feasible and bounded, then the dual problem is also feasible and bounded, and their optimal objective function values are equal. This is a fundamental result in linear programming duality.

   d. Incorrect. The dual problem is guaranteed to be bounded if the primal is feasible and bounded.

   e. Incorrect. Strong duality rules out both infeasibility and unboundedness for the dual if the primal is feasible and bounded.

7. Referring to Slide 2, explain the significance of the complementary slackness conditions $x_j z_j = 0$ and $y_i w_i = 0$ in relation to the optimal solutions of the primal and dual problems.

   a. Incorrect. While they are necessary, they are also sufficient for optimality given feasibility of both primal and dual.

   b. Incorrect. They are necessary conditions as well, meaning that if a solution is optimal, it must satisfy these conditions.

   c. Complementary slackness conditions are both necessary and sufficient for optimality when both the primal and dual problems are feasible. They state that at optimality, for each primal variable x_j, either $x_j = 0$ or its corresponding dual slack variable $z_j = 0$ (or both), and similarly for each dual variable y_i and its corresponding primal slack variable w_i.

   d. Incorrect. Complementary slackness is a fundamental concept in duality theory and crucial for determining optimality.

e. Incorrect. Complementary slackness applies to both bounded and unbounded problems, but its interpretation might differ slightly in unbounded cases.

8. Consider the graphical method illustrated in Slides 3 and 4. How does the optimal solution relate to the feasible region and the objective function?

   a. Incorrect. The optimal solution is not just any point; it must be a specific point on the boundary.

   b. Incorrect. This describes the optimal solution for a minimization problem, not a maximization problem.

   c. In the graphical method, the optimal solution for a maximization problem is always located at a vertex (or corner point) of the feasible region. This vertex is the point on the boundary of the feasible region that is furthest in the direction of improvement of the objective function.

   d. Incorrect. The centroid has no special significance in finding the optimal solution.

   e. Incorrect. The origin is only relevant if it is part of the feasible region.

9. Slides 11-15 discuss sensitivity analysis. If only the resource vector b changes in the primal problem (P), how does this affect the optimal objective function value ζ^*, and what is the economic interpretation of the vector y^* in this context?

   a. Incorrect. A change in b will generally change ζ^*, and y^* is the optimal dual solution, not primal.

   b. Incorrect. The change is not necessarily proportional, and y^* represents shadow prices, not reduced costs.

   c. The change in the optimal objective function value is given by $\Delta\zeta^* = (\Delta b)^T y^*$. The optimal dual variables y^* represent shadow prices, which are the marginal values of the resources. Each component of y^* indicates how much the optimal objective function value would increase if one more unit of the corresponding resource were available.

   d. Incorrect. The change is not necessarily inverse, and y^* represents shadow prices, not dual slack variables.

   e. Incorrect. The effect is predictable using the formula $\Delta\zeta^* = (\Delta b)^T y^*$, and y^* has a significant economic interpretation.

10. a. Given the primal problem (P) and its dual (D) as shown in Slide 1, if the primal problem is feasible and bounded, what can we conclude about the dual problem?

      A. This is incorrect. If the primal is feasible and bounded, the dual must also be feasible and bounded by the strong duality theorem.

      B. This is incorrect. If the primal is feasible and bounded, the dual cannot be unbounded.

      C. The strong duality theorem states that if the primal problem is feasible and bounded, then the dual problem is also feasible and bounded, and their optimal objective function values are equal.

      D. This is incorrect. The strong duality theorem provides a more precise conclusion.

      E. This is incorrect. The strong duality theorem provides a definitive conclusion.

    b. Referring to Slide 2, explain the significance of the complementary slackness conditions $x_j z_j = 0$ and $y_i w_i = 0$ in relation to the optimal solutions of the primal and dual problems.

A. While complementary slackness is related to the equality of primal and dual objective values at optimality, it doesn't directly ensure this equality. The equality is a consequence of strong duality.

B. Complementary slackness is a necessary condition for optimality, but it doesn't guarantee the existence of an optimal solution. Feasibility and boundedness are also required.

C. Complementary slackness states that for each primal variable x_j and its corresponding dual slack variable $z_j = (A^T y - c)_j$, the product $x_j z_j = 0$ at optimality. Similarly, for each dual variable y_i and its corresponding primal slack variable $w_i = (b - Ax)_i$, the product $y_i w_i = 0$ at optimality. This means that if a primal variable is positive, its corresponding dual slack variable must be zero, and vice versa. The same holds for dual variables and primal slack variables.

D. Complementary slackness, along with primal and dual feasibility, is a sufficient condition for optimality.

E. Complementary slackness applies to problems with both equality and inequality constraints.

c. Consider the graphical method illustrated in Slides 3 and 4. How does the optimal solution relate to the feasible region and the objective function?

A. This is incorrect; the optimal solution is a specific point, not any point within the feasible region.

B. This is incorrect for a maximization problem; it's true for a minimization problem.

C. In the graphical method, the feasible region is the set of all points satisfying the constraints. The objective function is represented by a line (or plane in higher dimensions). The optimal solution is the point in the feasible region that is furthest in the direction of improvement of the objective function. For maximization problems, this point is typically on the boundary of the feasible region, often at a vertex.

D. This is incorrect; the centroid has no special significance in linear programming.

E. This is incorrect; while the optimal solution is often at a vertex where multiple constraints are binding, it's not a requirement for all constraints to be binding.

d. Slides 11-15 discuss sensitivity analysis. If only the resource vector b changes in the primal problem (P), how does this affect the optimal objective function value ζ^*, and what is the economic interpretation of the vector y^* in this context?

A. The optimal objective function value does change if b changes. y^* represents shadow prices, not reduced costs.

B. If only the resource vector b changes, the change in the optimal objective function value $\Delta\zeta^*$ is given by $(\Delta b)^T y^*$. The optimal dual solution vector y^* represents the shadow prices, which are the marginal values of the resources. Each component y_i^* indicates how much the optimal objective function value would increase if one additional unit of resource i were available.

C. The change is proportional, not inverse. y^* represents shadow prices, not dual slack variables.

D. The change is predictable and given by the formula $(\Delta b)^T y^*$. y^* is the optimal dual solution, not primal slack variables.

E. The optimal objective function value changes if b changes. y^* is the optimal dual solution, not the optimal primal solution.